

Aryan Hemant Sahu

+91 8407968024 | aryansahu2705@gmail.com | Pune, India

 [Aryan Sahu](#) |  [aryaniscoding](#) | [LeetCode](#) | [Portfolio](#)

EDUCATION

• Pune Institute of Computer Technology (PICT)

B.E. in Electronics & Telecommunications (Honours in AI & Machine Learning)

CGPA: 9.604/10

2023 – Present

EXPERIENCE

• iMocha Technologies Pvt. Ltd.

GenAI Intern

Jan 2026 – Present

Hybrid

- Fine-tuning SOTA deep learning models (CNN, Transformer-based) for audio spoof detection in AI voice interview proctoring; evaluated on large-scale datasets using AUC-ROC and EER metrics.
- Architecting LLM-based agentic pipelines in n8n with prompt chaining and structured output parsing for automated Q&A generation, validation, and iterative refinement at scale.
- Engineered bulk-processing tools with schema validation and automation, cutting manual review by 60%.

• Shoption Pvt. Ltd.

IT Intern

Sep 2025 – Dec 2025

Hybrid

- Built a multilingual RAG backend (FastAPI, PostgreSQL + pgvector HNSW) using 768-dim bi-encoder embeddings with Gemini/Cohere LLMs for cross-lingual semantic retrieval (en/hi/mr) via batched neural machine translation.
- Implemented hybrid retrieval: dense ANN cosine search → BM25 full-text search → cross-encoder reranking (200→50→5 passage cascade) with real-time CRM enrichment via parallel async I/O.
- Instrumented end-to-end telemetry (inference latency, token usage, CPU/RAM); achieved 40% latency reduction via inference optimization and system-level architectural refactoring.

• NxTechWorks Consulting Pvt. Ltd.

Data Science Intern

May 2025 – Jun 2025

Remote

- Built a RAG system with bi-encoder retrieval and cross-encoder reranking (late interaction), improving top-k passage precision by 25%; integrated abstractive summarization with automatic QA-pair generation via instruction-tuned LLMs.
- Designed and deployed n8n workflows for personalized event recommendations via email and bidirectional Google Calendar sync.

• Aeronica Advance Technologies Pvt. Ltd.

Control Systems Intern

Jun 2025 – Aug 2025

Hybrid

- Implemented PID, MPC, and LQR algorithms on STM32 for tri-axis UAV stabilization with MPU6050 IMU feedback and Kalman filter-based state estimation.
- Reduced control loop response time by 13% through frequency tuning and latency, improving flight stability.

PROJECTS

• ContextMesh: Collaborative Team AI Context Network [MCP, NPM, React, Python](#)

Mar 2026 

- Architected a centralized AI context-sharing engine using Model Context Protocol (MCP) to synchronize persistent knowledge graphs across multi-agent toolchains (Claude Code, Antigravity) for coherent cross-session reasoning.
- Published an NPM CLI package automating MCP server provisioning, reducing distributed team IDE setup time by 80%; built a React dashboard compressing developer onboarding KT from 40+ hours to 4–5 hours.

• AquaMitra: Hybrid RAG + Text-to-SQL [Python, FastAPI, LlamaIndex, DuckDB, SQLAlchemy](#)

Aug 2025 

- Built a Hybrid RAG Q&A system with a LlamaIndex router dynamically switching between dense vector retrieval and Text-to-SQL inference over DuckDB based on query intent classification.
- Persisted chat history with executed SQL, retrieval scores, and per-query latency; exposed REST endpoints via FastAPI + Uvicorn with SQLAlchemy (duckdb-engine).

SKILLS

- **ML & Deep Learning:** CNNs, RNNs, LSTMs, Transformers, Attention Mechanisms, Fine-tuning, Transfer Learning, Self-supervised Learning, Contrastive Learning, OCR
- **NLP & GenAI:** RAG, Bi-encoder/Cross-encoder Reranking, Vector Embeddings, Prompt Engineering, LLM Inference, Instruction Tuning, Agentic AI, MCP, LangChain, LlamaIndex, N8N
- **MLOps & Infra:** Model Deployment, Docker, AWS, Git/GitHub, Hugging Face, Kaggle
- **Databases: Relational:** PostgreSQL, MySQL, DuckDB **Vector:** pgvector (HNSW, IVFFlat), FAISS
- **Programming & Frameworks:** Python, C/C++, FastAPI, Flask, Uvicorn, Pandas, NumPy, Scikit-learn, TensorFlow
- **CS Fundamentals:** DSA, OOP, DBMS, Computer Networks, System Design, Distributed Systems

ACHIEVEMENTS

- Solved 550+ DSA problems across LeetCode, CodeChef, and GfG, demonstrating strong algorithmic problem-solving ability
- Ranked in the **top 45 out of 650+** teams nationwide in the Dream11 Gameathon